DPUClock

1. Dynamic parameters:

TimeErrorPercent	Percentage of the duration of a simulation step. If the current simulation step begins with a time delay which is greater than (*TimeErrorPercent***Period*) a *TimeError* is reported. Thereby, Period [ms] is the duration of a simulation step as defined in the computer configuration (see "Reference SILAB Configuration and Operation"). Default: 3%.

2. Outputs:

Time	Current simulation time in milliseconds.
TimeError	Time delay in milliseconds.

3. Functional principle:

The DPUClock can be used to monitor the internal time constraints of DPUs on a certain computer. Time runs from the start of the simulation (beginning with 0 ms) and is increased according to the frequency of the computer the DPU runs on. If the computation of all DPUs of a computer takes longer than the clock cycle defined in the computer configuration, the *TimeError* increases.

If there is no delay (i.e. the current simulation step has started not later than *TimeErrorPercent* % of the defined period of the computer) the output *TimeError* shows a negative value. This value gives the remaining computation time of the last simulation step in milliseconds.

4. Example:

This is usually included in the SILAB base configuration.

```
DPUClock ~Clock
{
	Computer = { ALL };
	Index = 1;
};
```

Würzburger Institut für Verkehrswissenschaften GmbH
Handelsregister: HRB 6430 Sitz Würzburg
USt-IdNr.: DE198235088
Geschäftsführung: Alexandra Neukum

Robert-Bosch-Straße 4, D - 97209 Veitshöchheim
Telefon: +49 931 78009-0, Fax: +49 931 78009-150
www.wivw.de